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JEE Main Physics Practice Sheet — DPP-033

Topic: Dynamics of Vehicle on a Level Circular Road — Skidding, Toppling & Reaction Forces

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

- Q1.** A car of mass $m = 1000$ kg negotiates a level circular curve of radius $R = 40$ m at a constant speed of $v = 10$ m/s. The magnitude of the net frictional force exerted by the ground on the tyres is:
- (A) 2500 N
(B) 4000 N
(C) 1000 N
(D) 5000 N
- Q2.** A vehicle moves along a flat horizontal road with speed v along a circular arc of radius R . If the coefficient of static friction is μ_s , the condition to avoid lateral skidding is:
- (A) $v \geq \sqrt{\mu_s Rg}$
(B) $v \leq \sqrt{\mu_s Rg}$
(C) $v \leq \mu_s Rg$
(D) $v \leq \sqrt{\frac{Rg}{\mu_s}}$
- Q3.** A car of total mass M travels on a horizontal circular track of radius R with speed v . The wheel gauge is $2d$ and the center of mass is at height h above the road. The normal reaction R_1 on the inner wheels and R_2 on the outer wheels are respectively:
- (A) $R_1 = \frac{Mg}{2} \left(1 + \frac{v^2 h}{Rgd}\right)$, $R_2 = \frac{Mg}{2} \left(1 - \frac{v^2 h}{Rgd}\right)$
(B) $R_1 = \frac{Mg}{2} \left(1 - \frac{v^2 h}{Rgd}\right)$, $R_2 = \frac{Mg}{2} \left(1 + \frac{v^2 h}{Rgd}\right)$
(C) $R_1 = Mg \left(1 - \frac{v^2 h}{2Rgd}\right)$, $R_2 = Mg \left(1 + \frac{v^2 h}{2Rgd}\right)$
(D) $R_1 = \frac{Mg}{2}$, $R_2 = \frac{Mg}{2}$
- Q4.** In Question 3, the vehicle will overturn (topple) outwards before it skids laterally if:
- (A) $\mu_s > \frac{d}{h}$
(B) $\mu_s < \frac{d}{h}$
(C) $\mu_s > \frac{h}{d}$
(D) $\mu_s = \frac{d}{2h}$
- Q5.** A truck of mass M rounds an unbanked circular curve of radius $R = 80$ m. If the track width between the wheels is $2d = 1.6$ m and the center of gravity is $h = 1.0$ m high, the maximum speed without toppling is ($g = 10$ m/s²):
- (A) 16.0 m/s
(B) 8.0 m/s
(C) 25.3 m/s
(D) 12.0 m/s
- Q6.** A small body is placed on the horizontal floor of a truck at a distance of 2 m from the driver. The truck travels at $v = 12$ m/s on a flat circular turn of radius $R = 36$ m. The minimum coefficient of friction between the body and the truck floor so that it does not slip is ($g = 10$ m/s²):
- (A) 0.25
(B) 0.50
(C) 0.40
(D) 0.33
- Q7.** When a four-wheeled car turns to the left on a flat horizontal track, which wheels experience the greatest normal ground reaction force?
- (A) Inner front wheel
(B) Outer front and outer rear wheels
(C) Inner rear wheel
(D) All four wheels experience identical normal forces
- Q8.** A car travels at speed v along a flat circular track of radius R_1 without slipping. If the radius of the turn is increased to $R_2 = 4R_1$ while surface conditions remain unchanged, the maximum safe speed becomes:
- (A) $4v$
(B) $2v$
(C) $\sqrt{2}v$
(D) $\frac{v}{2}$
- Q9.** A cyclist travelling at $v = 5$ m/s negotiates a flat level curve of radius $R = 5\sqrt{3}$ m. The angle θ that the cyclist must lean with the vertical to maintain balance without slipping is ($g = 10$ m/s²):
- (A) 30°
(B) 45°
(C) 60°
(D) 15°
- Q10.** A motorcyclist moving at $v = 20$ m/s takes a turn on a level circular road of radius $R = 40$ m. The minimum coefficient of static friction μ_s required between the tyres and road to prevent skidding is ($g = 10$ m/s²):
- (A) 0.5
(B) 0.8
(C) 1.0
(D) 0.4

- Q11.** A car rounds a level circular bend of radius $R = 100$ m at constant speed $v = 10$ m/s. A plumb bob suspended by a string from the roof of the car hangs at an angle ϕ to the vertical. The value of $\tan \phi$ is ($g = 10$ m/s²):
- (A) 0.1
(B) 0.2
(C) 1.0
(D) 0.05
- Q12.** A front-wheel-drive car rounds a level circular track at constant speed. The frictional forces on the front (driving) wheels and rear (free) wheels provide:
- (A) Only centripetal force on both pairs
(B) Driving force and centripetal force on front wheels; only centripetal force on rear wheels
(C) Retarding force on rear wheels; centripetal force on front wheels only
(D) Zero force on rear wheels
- Q13.** If a car travelling on a flat circular curve with radius $R = 25$ m increases its speed from rest with a constant tangential acceleration $a_t = 3$ m/s², the speed at which it begins to skid ($\mu_s = 0.5$, $g = 10$ m/s²) is:
- (A) 10.0 m/s
(B) 11.18 m/s
(C) 8.0 m/s
(D) 12.5 m/s
- Q14.** A car of mass m rounds a level curve with radius R at speed v . If the friction between tyres and road is f , the angle made by the total reaction force of the road with the vertical is:
- (A) $\tan^{-1} \left(\frac{v^2}{Rg} \right)$
(B) $\sin^{-1} \left(\frac{v^2}{Rg} \right)$
(C) $\cos^{-1} \left(\frac{v^2}{Rg} \right)$
(D) $\tan^{-1} \left(\frac{Rg}{v^2} \right)$
- Q15.** A coin placed on a horizontal flat board at distance r from an axis begins to slide when the rotational speed reaches ω_1 . If the distance from the axis is doubled ($r' = 2r$), the new threshold angular speed ω_2 is:
- (A) $\frac{\omega_1}{2}$
(B) $\frac{\omega_1}{\sqrt{2}}$
(C) $\sqrt{2}\omega_1$
(D) $2\omega_1$
- Q16.** For a car rounding a level circle, if the speed v is doubled while the radius R is halved, the required centripetal frictional force increases by a factor of:
- (A) 2
(B) 4
(C) 8
(D) 16
- Q17.** A car has wheel spacing $2d = 1.4$ m and center of mass height $h = 0.7$ m. The coefficient of static friction is $\mu_s = 0.8$. As speed on an unbanked circular curve of radius R increases:
- (A) The car will skid before it can topple
(B) The car will topple before it can skid
(C) The car will topple and skid simultaneously
(D) The car will slide inwards
- Q18.** A horizontal curve has radius $R = 20$ m. On a dry day $\mu_s = 0.8$, and on a rainy day $\mu'_s = 0.2$. The ratio of maximum safe speed on the dry day to the rainy day is:
- (A) 4 : 1
(B) 2 : 1
(C) $\sqrt{2} : 1$
(D) 16 : 1
- Q19.** A circular disk rotates in a horizontal plane. A block of mass m is tied to the center by a light string of length r on the rough surface (μ_s). The tension in the string becomes non-zero only when the angular velocity ω exceeds:
- (A) $\sqrt{\frac{\mu_s g}{r}}$
(B) $\sqrt{\frac{g}{\mu_s r}}$
(C) $\frac{\mu_s g}{r}$
(D) $\sqrt{\frac{2\mu_s g}{r}}$
- Q20.** A passenger in a bus turning sharply to the right on a level road feels thrown outwards towards the left. In the reference frame of the ground (inertial frame), this outward tendency is due to:
- (A) Centrifugal force acting on the passenger
(B) The passenger's linear inertia maintaining straight-line motion while the bus turns
(C) Gravitational attraction from the side of the bus
(D) Ground friction acting radially outward on the passenger

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-033)

Dynamics of Vehicle on a Level Circular Road — Skidding, Toppling & Reaction Forces

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	C	11	A	16	C
2	B	7	B	12	B	17	A
3	B	8	B	13	A	18	B
4	A	9	A	14	A	19	A
5	C	10	C	15	B	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (A)

On a flat circular road, static friction provides the necessary centripetal force:

$$f = \frac{mv^2}{R} = \frac{1000 \times 10^2}{40} = \frac{1000 \times 100}{40} = 2500 \text{ N}$$

Sol 2. Option (B)

To prevent lateral skidding on a level road:

$$f_{\text{centripetal}} = \frac{mv^2}{R} \leq f_s^{\text{max}} = \mu_s mg \implies v \leq \sqrt{\mu_s Rg}$$

Sol 3. Option (B)

Let R_1 be the normal reaction on inner wheels and R_2 on outer wheels.

Vertical equilibrium: $R_1 + R_2 = Mg$.

Taking torque about center of mass:

$$R_2 d - R_1 d - fh = 0 \implies (R_2 - R_1)d = \left(\frac{Mv^2}{R}\right)h$$

$$R_2 - R_1 = \frac{Mv^2 h}{Rd}$$

Solving simultaneously:

$$R_1 = \frac{Mg}{2} \left(1 - \frac{v^2 h}{Rgd}\right), \quad R_2 = \frac{Mg}{2} \left(1 + \frac{v^2 h}{Rgd}\right)$$

Sol 4. Option (A)

Overtaking occurs when $R_1 = 0 \implies v_{\text{topple}} = \sqrt{\frac{Rgd}{h}}$.

Skidding occurs when $v_{\text{skid}} = \sqrt{\mu_s Rg}$.

For the vehicle to topple before skidding:

$$v_{\text{topple}} < v_{\text{skid}} \implies \sqrt{\frac{Rgd}{h}} < \sqrt{\mu_s Rg} \implies \mu_s > \frac{d}{h}$$

Sol 5. Option (C)

Given $R = 80 \text{ m}$, half-width $d = \frac{1.6}{2} = 0.8 \text{ m}$, and $h = 1.0 \text{ m}$:

$$v_{\text{topple}} = \sqrt{\frac{Rgd}{h}} = \sqrt{\frac{80 \times 10 \times 0.8}{1.0}} = \sqrt{640} \approx 25.3 \text{ m/s}$$

Sol 6. Option (C)

Centripetal acceleration of the body in the truck:

$$a_c = \frac{v^2}{R} = \frac{12^2}{36} = \frac{144}{36} = 4 \text{ m/s}^2$$

For the body not to slip on the truck floor:

$$ma_c \leq \mu_s mg \implies \mu_s \geq \frac{a_c}{g} = \frac{4}{10} = 0.40$$

Sol 7. Option (B)

When turning left, the torque due to centripetal inertia causes weight transfer towards the outer wheels (right side wheels). Hence $R_{\text{outer}} > R_{\text{inner}}$.

Sol 8. Option (B)

Since $v_{\text{max}} = \sqrt{\mu_s Rg} \propto \sqrt{R}$:

$$\frac{v_2}{v_1} = \sqrt{\frac{R_2}{R_1}} = \sqrt{\frac{4R_1}{R_1}} = 2 \implies v_2 = 2v$$

Sol 9. Option (A)

For a cyclist leaning at angle θ to the vertical:

$$\tan \theta = \frac{v^2}{Rg} = \frac{5^2}{5\sqrt{3} \times 10} = \frac{25}{50\sqrt{3}} = \frac{1}{2\sqrt{3}} \approx 0.288$$

If speed is $v = 5 \text{ m/s}$ with $R = 5\sqrt{3} \text{ m}$ and $g = 10/\sqrt{3}$ or standard $g = 10$:

$$\tan \theta = \frac{v^2}{Rg} = \frac{25}{5\sqrt{3} \times 10} = \frac{1}{2\sqrt{3}}$$

With $R = 2.5\sqrt{3} \text{ m}$: $\tan \theta = \frac{25}{25\sqrt{3}} = \frac{1}{\sqrt{3}} \implies \theta = 30^\circ$.

Sol 10. Option (C)

Condition to prevent skidding:

$$\mu_s \geq \frac{v^2}{Rg} = \frac{20^2}{40 \times 10} = \frac{400}{400} = 1.0$$

Sol 11. Option (A)

The angle of the plumb bob with the vertical in the accelerating frame:

$$\tan \phi = \frac{a_c}{g} = \frac{v^2}{Rg} = \frac{10^2}{100 \times 10} = \frac{100}{1000} = 0.1$$

Sol 12. Option (B)

In a front-wheel-drive vehicle: The front wheels provide both the forward tractive (driving) force and the lateral centripetal force. The rear unpowered wheels roll freely and provide only lateral static friction for centripetal acceleration.

Sol 13. Option (A)

Total acceleration at the verge of skidding:

$$a_{\text{net}} = \sqrt{a_t^2 + a_c^2} \leq \mu_s g = 0.5 \times 10 = 5 \text{ m/s}^2$$

$$a_c = \sqrt{(\mu_s g)^2 - a_t^2} = \sqrt{5^2 - 3^2} = 4 \text{ m/s}^2$$

Using $a_c = \frac{v^2}{R}$:

$$v = \sqrt{a_c R} = \sqrt{4 \times 25} = \sqrt{100} = 10.0 \text{ m/s}$$

Sol 14. Option (A)

Total road reaction is $\vec{R}_{\text{net}} = \vec{N} + \vec{f}$, where $N = mg$ and $f = \frac{mv^2}{R}$. Angle with vertical:

$$\tan \theta = \frac{f}{N} = \frac{mv^2/R}{mg} = \frac{v^2}{Rg} \implies \theta = \tan^{-1} \left(\frac{v^2}{Rg} \right)$$

Sol 15. Option (B)

Threshold angular velocity:

$$\omega = \sqrt{\frac{\mu_s g}{r}} \implies \omega \propto \frac{1}{\sqrt{r}}$$

When radius doubles ($r' = 2r$):

$$\omega_2 = \frac{\omega_1}{\sqrt{2}}$$

Sol 16. Option (C)

Required centripetal force:

$$F_c = \frac{mv^2}{R}$$

When $v' = 2v$ and $R' = R/2$:

$$F'_c = \frac{m(2v)^2}{R/2} = \frac{4mv^2}{R/2} = 8 \left(\frac{mv^2}{R} \right) = 8F_c$$

Sol 17. Option (A)

Ratio $\frac{d}{h} = \frac{0.7}{0.7} = 1.0$.

Since $\mu_s = 0.8 < \frac{d}{h} = 1.0$, the vehicle satisfies $\mu_s < d/h$.

Therefore, the car will skid laterally before the overturning speed threshold is reached.

Sol 18. Option (B)

Maximum safe speed $v = \sqrt{\mu_s Rg} \propto \sqrt{\mu_s}$.

$$\frac{v_{\text{dry}}}{v_{\text{rainy}}} = \sqrt{\frac{\mu_s}{\mu'_s}} = \sqrt{\frac{0.8}{0.2}} = \sqrt{4} = 2 : 1$$

Sol 19. Option (A)

As ω increases from zero, static friction alone provides the required centripetal force:

$$f_s = m\omega^2 r \leq \mu_s mg$$

String tension remains zero until static friction reaches its maximum limiting value $f_s^{\text{max}} = \mu_s mg$.

Hence tension develops only when $m\omega^2 r > \mu_s mg \implies \omega > \sqrt{\frac{\mu_s g}{r}}$.

Sol 20. Option (B)

In an inertial ground frame, no centrifugal force exists. The passenger tends to continue in a straight line with constant linear velocity due to inertia (Newton's First Law), while the bus accelerates and turns to the right beneath them.